

SAFETY CONCEPT DOCUMENT

Airbag Electronic Control Unit (ECU)

ISO 26262:2018 (ASIL D) / IEC 61508:2010 (SIL 3)

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1. Purpose and Scope

This Safety Concept Document (SCD) defines the strategy by which the Airbag Electronic Control Unit (ECU) achieves and maintains functional safety in accordance with ISO 26262:2018 and IEC 61508:2010. It translates the safety goals and hazardous events identified in the Hazard and Risk Analysis (HARA-AIRBAG-ECU-001 Rev 2.1) and the safety requirements specified in the Functional Safety Requirements Specification (FSRS-AIRBAG-ECU-001 Rev 2.1) into a coherent set of safety mechanisms, architectural decisions, fault-handling strategies, and safe-state definitions.

The Safety Concept answers three fundamental questions:

- How does the system prevent hazards from occurring?
- How does the system detect failures when they occur?
- What does the system do when a failure is detected (safe state)?

The document covers the confirmed laboratory implementation:

- Two Arduino Due boards in a dual-channel (2oo2) redundant architecture
- Three potentiometer-simulated analog crash sensors (POTI-1 accelerometer, POTI-2 gyroscope, POTI-3 pressure sensor)
- One digital baby-seat switch input
- CAN bus output to OMS laboratory PC with parallel bitstream readback

The mechanical airbag module, inflator, vehicle body structure, and pyrotechnic actuators are treated as adjacent systems outside the ECU item boundary.

2. Document Information and References

2.1 Reference Documents

Document	Document Number	Revision
Hazard and Risk Analysis	HARA-AIRBAG-ECU-001	Rev 2.1
Functional Safety Requirements Specification	FSRS-AIRBAG-ECU-001	Rev 2.1
ISO 26262:2018 — Road Vehicles Functional Safety	ISO 26262	2018 Edition
IEC 61508:2010 — Functional Safety of E/E Systems	IEC 61508	2010 Edition

2.2 Abbreviations

Abbreviation	Definition
ASIL	Automotive Safety Integrity Level
CAN	Controller Area Network
CRC	Cyclic Redundancy Check
DTC	Diagnostic Trouble Code
DWI	Diagnostic Warning Indicator
ECU	Electronic Control Unit
EMC	Electromagnetic Compatibility
FSRS	Functional Safety Requirements Specification
HARA	Hazard and Risk Analysis
MCU	Microcontroller Unit
OCS	Occupant Classification System
PMHF	Probabilistic Metric for Hardware random Failures
POST	Power-On Self-Test
SC	Safety Concept
SG	Safety Goal
SIL	Safety Integrity Level
SPFM	Single-Point Fault Metric
2oo2	Two-out-of-Two voting architecture

3. Safety Goals Summary

The following safety goals were derived from the HARA and govern all safety mechanisms defined in this document. The highest classification across all goals determines the overall item classification.

SG-ID	Safety Goal	ASIL	SIL	Governing Hazard
SG-01	The ECU shall NOT command deployment without a valid crash event	ASIL D	SIL 3	H1-01, H1-02, H1-04
SG-02	The ECU shall deploy airbags within the required timing window during valid crash events	ASIL D	SIL 3	H2-01, H2-02
SG-03	The ECU shall suppress passenger airbag when child seat or empty seat is detected	ASIL D	SIL 3	H1-04

SG-04	The ECU shall activate seatbelt pretensioners within required timing limits	ASIL C	SIL 2	H2-03, H2-04
SG-05	The ECU shall reject EMC disturbances and invalid electrical events	ASIL C	SIL 2	H1-05
SG-06	The ECU shall detect and report safety-critical faults via DWI within 800 ms	ASIL A	SIL 1	H4-01
SG-07	The ECU shall transmit crash notification CAN frames reliably within 15 ms of deployment	ASIL B	SIL 2	H4-02

Overall Item Classification: ASIL D (ISO 26262) | SIL 3 (IEC 61508)

4. Functional Safety Concept

The Functional Safety Concept describes the architectural and logical safety mechanisms that collectively ensure the system achieves all safety goals. The concept is structured around three pillars: prevention of unintended deployment, guaranteed deployment when required, and continuous fault detection.

4.1 Dual-Channel Redundant Architecture (2oo2)

The most critical architectural measure is the deployment of two independent Arduino Due microcontroller boards operating in a Two-out-of-Two (2oo2) voting logic configuration. This architecture is required to meet the ASIL D single-point fault metric (SPFM $\geq 99\%$) specified in ISO 26262.

Architecture principles:

- Channel A (Arduino Due #1) and Channel B (Arduino Due #2) independently read all sensor inputs, execute the crash detection algorithm, and calculate a deployment decision.
- A deployment output is issued if and only if both Channel A and Channel B independently and simultaneously arrive at the same deployment decision. This is the AND-gate deployment authorization logic.
- Both channels calculate an independent CRC over the CAN frame payload to verify data integrity across the communication interface.
- A cross-comparison interface allows the two channels to synchronize and verify each other's decision before any output is activated.
- Failure of either channel results in suppression of deployment for the affected channel, activation of the DWI, and logging of a DTC. The remaining channel continues monitoring.

This architecture ensures that no single hardware or software fault in either processing channel can independently cause an unintended airbag deployment (safety goal SG-01).

4.2 Deployment Authorization Logic

A deployment command shall only be issued when all of the following six conditions are simultaneously satisfied (FSR-03):

#	Condition	Safety Rationale
1	Crash sensor thresholds exceeded (at least one axis: accel, gyro, or pressure)	Prevents deployment in non-crash scenarios (road shock, minor impacts)
2	Channel A independently confirms crash detection	Eliminates single-point hardware fault in Channel A from causing deployment
3	Channel B independently confirms crash detection	Eliminates single-point hardware fault in Channel B from causing deployment
4	Both channels reach identical deployment decision (AND-gate agreement)	Prevents deployment from divergent channel states or communication faults
5	No critical fault condition exists in either channel	Prevents deployment during unsafe system states
6	Passenger suppression conditions are not active (baby-seat signal HIGH)	Prevents passenger airbag deployment with child seat or empty seat (SG-03)

The deployment authorization logic is implemented identically on both processing channels. Neither channel alone can independently trigger a deployment output.

4.3 Sensor Fusion Strategy

The crash detection algorithm fuses three analog sensor channels to form a deployment decision. In the laboratory implementation, these are represented by potentiometers on a breadboard:

Input	Real Sensor	Lab Substitute	Range	Threshold (deploy)
POTI-1	Accelerometer	Potentiometer 1	0–100 g	> 30 g (frontal)
POTI-2	Gyroscope / Angular Rate	Potentiometer 2	0–360 °/s	> 120 °/s (rollover)
POTI-3	Pressure Sensor	Potentiometer 3	0–500 kPa	> 200 kPa (impact)
Digital In	OCS / Baby-Seat Sensor	Digital Switch (active-LOW)	LOW / HIGH	LOW = suppress pax airbag

The algorithm evaluates a combination of sensor readings rather than a single sensor exceeding threshold, reducing the probability of spurious deployment from vibration, road shock, or a single faulty potentiometer channel.

4.4 Occupant Classification and Passenger Airbag Suppression

The passenger airbag suppression logic is a dedicated ASIL D safety function addressing Safety Goal SG-03. It manages the baby-seat digital input as follows:

- Baby-seat input LOW (0 V): baby seat or child seat detected — passenger airbag is unconditionally suppressed.
- Baby-seat input HIGH (3.3 V): adult occupant or empty seat — passenger airbag remains armed.
- Suppress-on-fault strategy: If the baby-seat signal is unavailable, intermittent, or invalid, the ECU defaults to passenger airbag suppression. This is the fail-safe direction — suppression is safer than unintended deployment into a child seat.
- Signal debounce: The baby-seat digital input must remain stable for at least 10 ms before any state transition is accepted as valid (FSR-11).
- Hardware pull-up: A 10 kΩ pull-up resistor to 3.3 V ensures that an open-circuit wire condition defaults the input to HIGH. When combined with the suppress-on-fault software strategy (which also suppresses on unstable signals), this ensures that any open-circuit fault always results in suppression.
- Periodic line continuity check: The ECU drives the line LOW and verifies the response to detect stuck-HIGH (open-circuit) faults before a crash event occurs.

4.5 Hardware-Enforced Deployment Inhibit

Independent of software execution, a hardware safety switch (firing inhibit path) directly controls all firing transistors. This hardware path is not controlled by software and ensures that:

- No software runaway, memory corruption, or runaway thread can independently cause airbag deployment.
- The hardware inhibit remains active unless both processing channels have explicitly authorized deployment AND the hardware safety switch has received a verified enable signal.
- In case of system shutdown (SS-05), the hardware safety switch forces all output transistors off, independent of MCU state.

This satisfies Safety Constraint SC-09: Deployment inhibit functionality shall remain enforceable by hardware independent of software execution.

5. Fault Detection Mechanisms

Fault detection is implemented through two complementary strategies: (1) Power-On Self-Test (POST) at startup, and (2) Continuous Background Diagnostics during operation.

5.1 Power-On Self-Test (POST)

At every ignition cycle, the ECU completes a full POST sequence within 800 ms before arming the system (FSR-17). The POST verifies:

POST Item	Method	Action on Failure
MCU CPU core integrity	Lockstep core checkerboard test	System shutdown; DWI ON; DTC logged
Flash memory integrity	CRC32 check of code and calibration regions	Deployment inhibited; DWI ON; DTC logged
RAM integrity	Write/read pattern; stack boundary check	System reset; if recurring — shutdown
ADC / sensor plausibility	All three POTIs read within valid 0–3.3 V ADC range	Affected sensor channel flagged; DWI ON
Clock monitor / watchdog	Watchdog self-test; PLL lock verification	Watchdog-triggered reset; DWI ON
CAN bus communication	Test frame sent to OMS lab PC; receipt confirmed; CRC verified	CAN fault DTC logged; DWI ON; limp-home mode
Baby-seat input continuity	Line driven LOW; response verified	Suppress-on-fault applied; DWI ON; DTC logged
Power supply voltage	Analog supervisor: 9 V – 16 V window check	Low: hold + deploy from capacitor; High: firing inhibit
DWI lamp circuit	Drive current monitoring at power-on	DWI fault DTC; BCM notified via CAN

If any POST item fails, the DWI lamp is illuminated, a DTC is stored, and the system enters a safe degraded state as appropriate to the failure. Deployment is not armed until the POST passes.

5.2 Continuous Background Diagnostics

During normal operation, the ECU executes background diagnostics at intervals not exceeding 100 ms (FSR-18). These include:

- Sensor plausibility monitoring: All three analog potentiometer channels are monitored for values outside the valid ADC range (open-circuit, short, stuck-at-rail). Fault triggers DWI and DTC.
- Lockstep MCU core comparison: Channel A and Channel B compare their intermediate computation results every cycle (≤ 1 ms). Any mismatch triggers an immediate watchdog reset. If the mismatch persists after three consecutive resets, the system enters SS-05 (system shutdown).
- CAN frame timeout supervision: If no valid CAN frame is received from the opposing channel for more than 50 ms, a timeout fault is declared (FSR-15). The deployment algorithm continues using the last valid occupant classification, and pretensioners are not suppressed.
- Watchdog timer supervision: An independent hardware watchdog (timeout < 10 ms) monitors software execution. Failure to kick the watchdog within the timeout interval triggers an MCU reset. This detects software lockup, infinite loops, and runaway threads.
- Memory integrity monitoring: RAM CRC and stack boundary monitoring execute continuously. Detection of corruption triggers safe-state transition (FSR-22).
- Temperature monitoring: An NTC sensor on the ECU PCB is read continuously. If temperature exceeds 90 °C, a warning DTC is logged. If temperature exceeds 105 °C, deployment is inhibited to prevent inflator malfunctions.

- Squib loop integrity: All squib circuit resistances are monitored continuously within the valid window (1.5 Ω – 4.0 Ω). Open-circuit, short-to-battery, or short-to-ground on any channel triggers DWI and disables the affected channel while leaving the rest armed.

5.3 Fault Detection Summary Table

Failure Mode	Detection Method	Safe State	ASIL
MCU internal fault	Lockstep core comparison	SS-05 (system shutdown)	ASIL D
Sensor out-of-range	ADC range check, POST and background	SS-04 (channel isolated)	ASIL D
Baby-seat line fault (stuck-HIGH)	Continuity check, debounce	SS-03 (pax airbag suppressed)	ASIL D
Baby-seat line fault (stuck-LOW)	Signal monitoring, DWI	SS-03 + DWI ON, DTC	ASIL C
Flash memory CRC error	CRC32 at POST and periodically	SS-05 (deployment inhibited)	ASIL D
Software watchdog timeout	Hardware watchdog (< 10 ms)	MCU reset; SS-05 if recurring	ASIL D
CAN bus communication loss	Frame timeout (> 50 ms)	Limp-home, no OCS suppression	ASIL B
Power supply undervoltage	Analog voltage supervisor (< 9 V)	Backup capacitor; deploy still possible	ASIL C
Power supply overvoltage	Analog clamp + interrupt (> 16 V)	SS-01 (firing transistors inhibited)	ASIL C
Overtemperature	NTC on PCB (> 90 °C warn; > 105 °C inhibit)	SS-01 at > 105 °C	ASIL C
Squib loop open/short	Resistance window monitor (1.5–4.0 Ω)	SS-04 (affected channel only)	ASIL D
DWI lamp circuit fault	Drive current monitoring at POST	BCM CAN notification; DTC logged	ASIL A

6. Safe State Definitions

ISO 26262 requires that each safety goal has an associated safe state — the system condition that must be achieved or maintained when a fault is detected. The Airbag ECU defines five safe states:

SS-ID	Name	System Condition	Applicable Hazard(s)
SS-01	Deployment Inhibit State	All firing transistors remain open (OFF). Reservoir capacitor maintained but no	H1-01, H1-02, H1-03, H1-05

		firing command issued. DWI lamp ON if fault detected.	
SS-02	Deployment Armed State	Normal monitoring standby. All squib channels armed. Crash detection algorithm running continuously. Any fault disables only the affected channel — not the full system.	H2-01, H2-02, H2-03
SS-03	Passenger Airbag Suppressed	Passenger airbag firing transistor inhibited. Driver and side airbags remain armed. OCS fault or baby-seat signal loss triggers this state. DWI ON.	H1-04
SS-04	Partial Degradation State	Faulted squib channel or sensor channel isolated and disabled. Remaining channels continue normal operation. DWI lamp ON. DTC logged.	H3-01, H3-02, H4-01
SS-05	System Shutdown State	All output transistors forced OFF by hardware safety switch, independent of MCU state. Watchdog asserts reset. If fault persists after 3 resets — permanent shutdown with DWI lamp on. Cannot be reset without authorised OBD workshop procedure.	H1-01, H2-01 (related)

State Transition Principle:

- All transitions into a safe state are irreversible without an authorised reset (OBD-II workshop procedure) for SS-05.
- Transitions from SS-02 (armed) to SS-01 or SS-03 occur automatically upon fault detection within one diagnostic cycle (≤ 100 ms).
- The DWI lamp is activated within 800 ms of any fault that reduces the system below full protection capability (FSR-19).

7. Safety Mechanisms

7.1 Watchdog Timers

Two independent watchdog supervisors are implemented:

- **Hardware watchdog (independent of MCU):** The primary watchdog is implemented in hardware, outside the MCU, and cannot be disabled by software. It must be regularly kicked by the running software within a 10 ms timeout window. A watchdog timeout indicates software deadlock, runaway, or crash — triggering an MCU reset and transitioning to the safe state.
- **Software watchdog (secondary):** A software-implemented task monitor verifies that safety-critical tasks complete within their allowed time windows. Non-completion triggers a controlled safe-state

transition before the hardware watchdog fires, enabling a more graceful response with DTC logging.

7.2 Cross-Channel Comparison

Channel A and Channel B compare their deployment decisions and intermediate computational results before issuing any output. This cross-channel comparison mechanism:

- Detects hardware faults that cause one channel to produce incorrect sensor readings or deployment decisions
- Detects software faults that cause divergent algorithm execution between the two channels
- Detects communication faults that corrupt the cross-channel decision synchronization interface
- Any detected discrepancy between channels results in deployment being held and a fault being logged

The cross-comparison executes at every 1 ms algorithm cycle, satisfying the FSR-02 deterministic cycle time requirement.

7.3 CRC Protection

CRC protection is applied at three levels:

- Flash memory integrity: A CRC32 is calculated over the entire program code and calibration data at POST and periodically during operation (FSR-22). Corruption inhibits deployment.
- CAN frame integrity: Every CAN frame transmitted to the OMS lab PC includes a 16-bit CRC field calculated by both channels using the Arduino CRC library (FSR-13). Frames with invalid CRC are discarded.
- RAM integrity: Runtime RAM regions are protected by CRC verification. Stack boundary monitoring prevents overflow conditions from corrupting safety-critical variables (FSR-22).

7.4 Parallel CAN Bitstream Readback

A secondary parallel readback channel monitors the transmitted CAN bitstream directly after transmission (FSR-14). This detects:

- CAN transceiver hardware faults causing bit errors during transmission
- Corrupted CAN frames not detected by the standard CAN ACK mechanism
- Differences between the intended and transmitted bitstream

Any detected discrepancy between the transmitted frame and the readback channel result triggers a CAN fault DTC and activates the DWI.

7.5 EMC Protection

To satisfy Safety Goal SG-05 (EMC disturbance rejection, ASIL C), the following hardware and software measures are implemented:

- Galvanic isolation on squib firing lines — prevents radiated disturbances from directly activating the firing circuit.
- Differential signal lines and shielded squib wiring — reduces susceptibility to conducted and radiated EMC events.

- **Hardware firing key** — a separate hardware enable signal must be asserted by the MCU in addition to the firing transistor gate control. An EMC-induced gate voltage cannot deploy the airbag without the hardware key.
- **Software plausibility check** — all three sensor channels must exceed threshold simultaneously before deployment is authorized. An EMC-induced spike on a single sensor channel cannot alone cause deployment.
- EMC testing per CISPR 25 and ISO 11452 series is required during design validation.

7.6 Memory Safety

Memory safety mechanisms implement FSR-22 and protect against software runaway, stack overflow, and memory corruption:

- **RAM integrity:** CRC or equivalent technique applied to safety-critical RAM regions at each diagnostic cycle.
- **Stack boundary monitoring:** A guard region at the stack boundary is monitored. Access to the guard region triggers an immediate safe-state transition.
- **Invalid memory access detection:** The MCU memory protection unit (MPU) flags unauthorized memory accesses and triggers a fault handler.
- **Watchdog supervision:** The hardware watchdog prevents runaway software execution from persisting beyond 10 ms.
- **Safe-state transition:** Upon detection of any unrecoverable memory fault, all firing outputs are disabled and the system enters SS-05.

7.7 Timing Supervision

Deployment timing is a safety-critical parameter. The following timing constraints are monitored:

Function	Maximum Time	Supervision Method
Crash detection algorithm cycle	1 ms	Software task monitor; hardware watchdog
Crash detection response (threshold to decision)	20 ms	Algorithm deadline monitoring
Deployment command transmission	40 ms	Timestamp comparison post-event
Side-impact deployment logic	< 8 ms	Dedicated high-priority interrupt path
Pretensioner activation	< 5 ms	Sequenced simultaneously with deployment command
CAN timeout detection	50 ms	CAN frame timestamp; timeout interrupt
POST execution (complete before arming)	800 ms	Startup sequence controller
DWI activation after fault detection	800 ms	Fault handler invocation time
Baby-seat debounce	10 ms minimum stable	Digital input filter counter

8. Safety Mechanism Traceability

The following matrix maps each safety mechanism to the safety goals it contributes to, ensuring complete coverage of all ASIL D / SIL 3 requirements.

Safety Mechanism	SG-01	SG-02	SG-03	SG-04	SG-05	SG-06	SG-07
Dual-channel 2oo2 architecture	✓	✓					
AND-gate deployment authorization (6-condition)	✓	✓	✓				
Hardware-enforced firing inhibit	✓				✓		
Baby-seat suppress-on-fault (default suppress)	✓		✓				
Hardware pull-up + debounce filter			✓				
Lockstep MCU cross-channel comparison	✓	✓				✓	
Hardware watchdog timer (< 10 ms)	✓	✓		✓			
CRC protection (Flash, RAM, CAN frames)	✓	✓				✓	✓
Parallel CAN bitstream readback							✓
Power-On Self-Test (POST)	✓	✓	✓			✓	
Continuous background diagnostics (≤ 100 ms)	✓	✓	✓	✓		✓	
DWI lamp activation (≤ 800 ms)						✓	
EMC hardware measures (isolation, shielding, key)	✓				✓		
Memory protection (MPU, stack guard, RAM CRC)	✓	✓				✓	
Safe-state transitions (SS-01 through SS-05)	✓	✓	✓	✓	✓	✓	✓

9. Safety Constraints

The following safety constraints are derived from the safety concept and must be satisfied by all hardware and software implementations. They are non-negotiable and cannot be relaxed without a full re-assessment of the HARA and ASIL classification.

SC-ID	Safety Constraint	ASIL / SIL	Source
SC-01	Overall system classification shall remain ASIL D / SIL 3	ASIL D / SIL 3	HARA
SC-02	PMHF shall remain below 10^{-8} failures/hour	ASIL D / SIL 3	ISO 26262
SC-03	No single-point fault shall cause unintended airbag deployment	ASIL D / SIL 3	SG-01
SC-04	Deployment requires agreement of both redundant channels	ASIL D / SIL 3	FSR-08
SC-05	All CAN communication shall include CRC verification	ASIL B / SIL 2	FSR-13
SC-06	Detection of any critical fault shall force transition to a defined safe state	ASIL D / SIL 3	FSR-20
SC-07	Safety-related software shall execute deterministically within defined time limits	ASIL D / SIL 3	FSR-02
SC-08	Diagnostic monitoring shall execute continuously during all operational states	ASIL C / SIL 2	FSR-18
SC-09	Deployment inhibit functionality shall remain enforceable by hardware independent of software execution	ASIL D / SIL 3	FSRS SC-09
SC-10	Baby-seat signal loss or instability shall default to passenger airbag suppression (suppress-on-fault)	ASIL D / SIL 3	FSR-10
SC-11	SPFM for ASIL D functions shall be $\geq 99\%$ as required by ISO 26262 Part 5	ASIL D / SIL 3	ISO 26262-5

10. Revision History

Rev	Date	Author	Change Description
1.0	2026-05-06	Systems Safety Engineering Team	Initial release- Full Safety Concept Document
2.0	2026-05-12	Systems Safety Engineering Team	Updated release. Full Safety Concept covering Functional Safety Concept, Fault Detection strategy, Safe State definitions, Safety Mechanisms (watchdog, CRC, cross-channel, EMC, memory safety), and complete safety mechanism traceability matrix.